

Review: Comparative Transcriptomics of Adherent and Suspension Chicken Fibroblast Cell Lines for the Optimization of Cultivated Meat Processes

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Abstract:

Global meat demand is expected to continue rising, and because poultry is one of the most widely consumed meats worldwide, developing alternatives to conventional chicken farming could help mitigate its environmental, animal welfare, and public health impacts. Cultivated meat is a potential solution, but scaling remains challenging because many animal cells require attachment to scaffold materials, which reduces yield and increases cost. Developing non-adherent or suspension cell lines would eliminate the need for scaffolds, enabling higher cell densities and simplifying harvesting, making suspension lines a promising strategy for commercializing cellular agriculture.

This review focuses on a study by Contreras et al. (2026), which adapted a commercially available chicken embryonic fibroblast cell line (DF-1) to suspension culture using a previously published method by Pasitka et al. (2023). The study aimed to address the limited understanding of the cellular and molecular changes that occur during the transition from adherent to non-adherent growth states through RNA sequencing methods. This review summarizes this suspension adaptation process, the proposed mechanism, and highlights the key challenges that must be addressed for its successful application in commercial cellular agriculture.

Introduction:

Most muscle, fat, and other food-relevant cell types are inherently adherent due to the presence of integrins, which mediate attachment to the extracellular matrix (ECM). These interactions are essential for proper cell function within a living organism (Danen, 2013). When

integrin-ECM interactions are disrupted, the organization of the actin cytoskeleton is altered, and cells may undergo anoikis, a form of caspase-mediated programmed cell death (Malagobadan & Nagoor, 2019). Therefore, successful adaptation to suspension culture partly requires the development of anoikis resistance.

Anoikis-resistant cells can arise through several mechanisms. (1) Reactive oxygen species and kinase activation can stimulate survival pathways that typically depend on integrin signaling. (2) Hypoxia can stabilize hypoxia-inducible factor 1 (HIF-1) transcription factors, which activate genes that enable cells to adapt to metabolic stress (Whelan et al., 2010). And (3), oncogene activation, mutations that drive uncontrolled cell growth, can suppress apoptotic pathways and promote cell survival (Cooper, 2000).

However, overcoming anoikis alone is not sufficient for successful suspension growth. Cells that survive detachment may still struggle to proliferate efficiently, divide normally, or maintain long-term stability. Because these cells evolved to function in an adherent environment, their metabolism and regulatory pathways are not inherently optimized for suspension culture.

There are two major adaptation methods: non-targeted, “spontaneous,” selection and genetic engineering. There is a lack of a clear mechanistic understanding of how mammalian cells proliferate under suspended conditions for both approaches. Additionally, spontaneous selection is inefficient, prone to high failure rates, and difficult to reproduce, while genetic engineering can introduce unpredictable downstream effects.

The study that originally described the protocol used by Contreras et al. (2026) primarily focused on cell line immortalization rather than suspension adaptation. In that work, RNA sequencing was used to compare primary adherent cells with immortalized cells that happened to exhibit suspension growth capability. This design confounded whether the observed gene and

pathway enrichment was driven by anchorage independence or by immortalization. In contrast, Contreras et al. (2026) isolated suspension adaptation as the primary variable, enabling a clearer analysis of the transcriptional mechanisms underlying anchorage-independent growth.

Methods of Adaptation:

The selected protocol involves five stages (T1-T5), allowing cells to gradually transition to suspension culture through selection pressure that enriches for anoikis-resistant, non-adherent, proliferative cells in serum-supplemented media. Four independent replicates were run in parallel.

In T1, cells were grown in standard adherent culture conditions and displayed the typical elongated, spindle-shaped fibroblastic morphology. In T2, the cells were transferred to AggreWell plates with conical microwells for 24 hours, where the T1 cells aggregated into uniform anchorage-independent spheroids approximately 15 times the diameter of the individual T1 cells. In T3, the spheroids were moved into a static suspension culture for four days. During this stage, the spheroids were mechanically redistributed through daily micropipetting, causing the cells to aggregate into larger, irregular clusters.

In T4, the clusters were transferred to dynamic suspension culture at 80 rpm for nine days, which reduced aggregate size. Finally, in T5, agitation was increased to 100 rpm, further breaking the clusters into smaller aggregates and single cells. This two-step increase in rpm was designed to allow cells to gradually adapt to increasing hydrodynamic shear stress while improving nutrient and oxygen transfer.

End of Adaptation Cycle:

Cells at T5 were considered suspension-adapted based on their stable growth characteristics, including an average doubling time approximately 1.6 times longer than the

original adherent cells and an average cell size approximately 94% of the original. The longer doubling time and smaller average size suggest a less efficient metabolism or other deficiencies due to lack of adherence.

Despite these changes, the relative viable cell yield (cells produced per mL of media) was higher in suspension cultures than in anchorage-dependent cultures. This occurs because suspension growth is limited by culture volume rather than surface area, allowing for a higher achievable viable cell density (VCD). The percentage of viable cells after passing into fresh media was similar for both adherent and suspension cultures, suggesting that cells were dying at comparable rates. So the suspension capable cells improve the overall yield because cells can exist in the same volume, not because they reproduce faster.

The authors hypothesized that certain transcriptomic differences drive the phenotypic changes observed between adherent and suspension cells. To investigate this, RNA was collected from cells at each stage of the adaptation pipeline (T1-T5) for transcriptomic analysis.

Trajectory Principal Component Analysis (PCA):

PCA identifies principal components by determining the directions in the dataset that capture the greatest variance in gene expression across samples. Interpretation showed that the aggregate size between samples accounted for 56% of the variation in gene expression. The differences in the magnitude of mechanical shear stress experienced by the cells accounted for 29% of the variation. Together, these results suggest that aggregate size and hydrodynamic shear stress are major drivers of transcriptomic changes during the suspension adaptation process. Additionally, PCA showed strong consistency between the four biological replicates, with tight clustering of samples within each stage of the adaptation pipeline – suggesting minimal batch

effects and distinct transcriptomic changes throughout the suspension adaptation process (“Principal Component Analysis,” 2026).

Differential Gene Expression Analysis (DGEA):

DGEA was used to determine which genes changed their expression levels between conditions relative to the T1 adherent baseline (What Is Differential Gene Expression Analysis? - CD Genomics, n.d.). The study applied a 1.5-fold change threshold and a 5% Benjamini-Hochberg false discovery rate (FDR); the latter estimates the proportion of significant results that are likely to be false positives (False Discovery Rate, 2016). This correction is necessary because when thousands of genes are tested simultaneously, a large number of genes may appear statistically significant purely by chance.

The results showed that T3 exhibited the largest number of gene expression changes relative to the T1 baseline, while T2, T4, and T5 showed fewer changes. This suggests that T3 represents the peak transition stage of the adaptation process. The authors guessed that this is where cells downregulate genes associated with adherent growth while upregulating genes that support survival and proliferation in suspension, meaning T3 may represent the point at which cells commit to the suspension phenotype.

Late-stage transcriptional changes suggested stabilization of the suspension phenotype. Although PCA suggested that T4 and T5 samples separated due to differences in shear stress, DGEA revealed only a 0.22% difference in expressed genes between these stages. This suggests that the transcriptome had largely stabilized by T4, and that the increased agitation between T4 and T5 did not substantially alter gene expression. Because the agitation speeds used in these stages are similar to those estimated for cultivated meat bioreactors, the authors suggest that

suspension-adapted DF-1 cells may remain stable under industrial production conditions. It is unclear however how close the Reynolds numbers, a better metric for process scaling, are.

Upset Analysis:

UpSet analysis was then used to identify intersections between sets of differentially expressed genes across stages (Lex et al., 2014). The largest intersection consisted of genes shared between T2 and T3 relative to T1, including 477 upregulated genes and 270 downregulated genes. The upregulated genes were enriched in pathways related to signal transduction and cell motility, suggesting that cells adopt a more migratory and communicative state during the early stages of suspension adaptation. The downregulated genes were associated with ribosomal biogenesis, one of the most resource intensive processes, which may contribute to the smaller average size observed in non-adherent DF-1 cells.

Gene Set Enrichment Analysis (GSEA):

DGEA identified individual genes that change between conditions, and then GSEA was used to evaluate whether groups of genes belonging to specific biological pathways are collectively upregulated or downregulated compared to T1 (Subramanian et al., 2005). GSEA was performed for each contrast using the Kyoto Encyclopedia of Genes and Genomes (KEGG) database to identify pathways associated with the observed transcriptional changes.

GSEA results suggest that transcriptional activity becomes more stable over time. Early in the adaptation process (T2) there are 45 differentially enriched pathways (DEPs), whereas the final stage (T5) had a very similar transcriptional and functional profile to T1 with only 7 DEPs. This pattern is consistent with a process of acute stress response followed by stabilization as cells adapt to suspension conditions. In the acute stress response (T2 and T3), cells temporarily reduce proliferation-related processes. Apoptosis and autophagy are upregulated. DNA replication and

repair are suppressed. Pathways involved in cell junctions, including focal adhesion and ECM-receptor interactions, were also activated – likely reflecting attempts by the cells to compensate for the loss of attachment by increasing the production of cell adhesion molecules (CAMs) and reorganizing adhesion signaling pathways (McNiven & Razidlo, 2016).

Although the number of (DEPs) decreased over time, cellular morphology continued to change throughout the adaptation process. During the earlier stages (T2 and T3), ATP-generating processes such as oxidative phosphorylation and amino acid degradation pathways were repressed – explaining the observed reduction in transcriptional activity as cells likely redirected energy toward stress response mechanisms rather than growth. By T4, additional pathways involved in cofactor biosynthesis, nucleocytoplasmic transport, and proteasome function were suppressed, suggesting a temporary reduction in both protein synthesis and protein degradation during adaptation. By T5, however, many of these suppressed pathways had returned toward baseline levels, suggesting recovery of normal transcriptional and translational activity.

Six pathways appeared in all four set comparisons (T2-T5 relative to T1). Hormone signaling, cytokine-cytokine receptor interaction, and lysosome pathways were upregulated – suggesting that suspension-adapted cells rely more heavily on intercellular communication and stress-response signaling to maintain homeostasis and adapt to their new growth environment. Steroid biosynthesis, ribosome biogenesis, and spliceosome pathways were downregulated, likely reducing the capacity for protein synthesis, RNA processing, and lipid metabolism. This suggests that suspension-adapted cells operate in a lower metabolic state compared with adherent cells, possibly as a strategy to conserve energy under altered growth conditions.

Mfuzz Analysis:

Mfuzz was used to analyze gene expression changes over time rather than only differences between experimental stages by clustering genes with similar expression trajectories across the time course (Futschik & Kumar, 2024). Mfuzz uses soft clustering, meaning that a gene can belong to multiple clusters with different membership probabilities, better reflecting biological reality since genes often participate in multiple regulatory programs or pathways.

To interpret the biological significance of these clusters, the study performed functional enrichment analysis (FEA) on the six identified Mfuzz clusters to determine whether specific biological processes were overrepresented compared to random expectation. This analysis used the KEGG and Gene Ontology (GO) databases. There were six different Mfuzz clusters.

At T3, Mfuzz expression pattern clusters 1 and 5 displayed expression maxima while cluster 4 showed a minimum, all followed by stabilization at later stages. This pattern supports previous DGEA and GSEA results suggesting that T3 represents the most stressful stage of the adaptation pipeline.

FEA suggested that Mfuzz clusters 1 and 5 were enriched for stress-response pathways, including autophagy and apoptosis. Because these expression changes peaked at T3, when cell aggregates exceeded the ~200 μm diameter limit at which oxygen diffusion typically becomes limiting, the authors suggest that hypoxia in the centers of aggregates may activate these pathways. In contrast, at T3, cluster 4 had suppression of pathways related to DNA replication and ribosome biogenesis, suggesting temporary suppression of growth-related processes during hypoxic stress.

Cluster 6 showed a spike in expression at T2 followed by downregulation at later stages. During T2, FEA showed that cells overexpress genes involved in cell junction pathways,

particularly focal adhesion and ECM-receptor interaction pathways – likely reflecting attempts by the cells to compensate for the loss of substrate attachment, which is also supported by previous GSEA. GO enrichment analysis also showed activation of vasculature and angiogenesis-related pathways at T2, which may represent signaling responses to emerging hypoxia (Adair & Montani, 2010).

FEA showed that lipid, steroid biosynthesis, and focal adhesion pathways were highly represented in cluster 2, which exhibited consistent downregulation. This finding supports previous DGEA and GSEA results suggesting reduced lipid metabolism. Transcription factors involved in regulating lipid metabolism and adipocyte differentiation were also present in this cluster, which the authors suggest means DF-1's may retain potential for transdifferentiation into adipocytes, which can be valuable for producing realistic meat products. However, because these transcription factors are downregulated, the functional relevance of this potential is unclear.

Cluster 3 showed consistent upregulation, but the GO and KEGG databases were unable to assign these genes to a shared pathway. This does not necessarily mean that the genes are unrelated; rather, they may not belong to currently defined pathways, or the databases may lack sufficient annotation for them. To address this, the authors manually examined the genes within the cluster – revealing several genes involved in extracellular matrix remodeling, tissue structure changes, and cytoskeletal and CAM reorganization. Additionally, genes known to induce p53-dependent apoptosis were present in cluster 3, suggesting that cells that cannot handle the stress well will experience p53 activation and undergo apoptosis during the adaptation process, selecting for a more stable population capable of surviving in suspension culture.

Quantitative PCR (qPCR) Confirmation:

Previous RNA sequencing analysis identified genes whose expression changed between conditions and quantified the magnitude of those changes. These results were validated using qPCR to measure endpoint gene expression differences between adherent cells (T1) and cells adapted to suspension conditions (T5) for a subset of eleven genes – chosen based on their adjusted p-values in the T1 vs. T5 contrast, because they exhibited at least a two-fold change in expression, or due to their relevance to cell adhesion. The results suggested strong agreement between qPCR and RNA sequencing, with a very low probability that this relationship occurred by chance. Therefore, the gene expression changes identified by RNA sequencing are considered reliable.

Conclusion: The authors propose that at T2 and T3, stress-activated p38 δ kinase sequesters TEAD family transcription factors in the cell's cytoplasm, thereby preventing their interaction with YAP (Yes-associated protein) (O'Callaghan et al., 2014; Zhou et al., 2016; Wu et al., 2019). As a result, YAP can no longer drive TEAD-dependent adhesion-related gene expression. Instead, YAP shifts its binding partner to FoxO1, a stress-response transcription factor, activating antioxidant and survival pathways (Yan et al., 2025). This switch simultaneously reduces cell-matrix adhesion signaling and enhances resistance to oxidative stress, enabling cells to detach from the extracellular matrix while avoiding anoikis.

The authors pointed out the lack of qPCR validation for the transient responses observed at T2 and T3, where the largest transcriptional changes occur and key adaptation mechanisms are proposed to happen. Validating these stages would strengthen confidence that the RNA sequencing results reflect true biological changes rather than transient or method-specific effects.

Additionally, it was noted that cells that progressed through the full adaptation pipeline and those adapted directly from T1 to T5 exhibited similar endpoint gene expression profiles. This suggests that the intermediate stages (T2-T4) may not be strictly required for achieving the suspension phenotype and could potentially be bypassed or accelerated. However, because the proposed mechanism emphasizes these stages as critical transition points driven by oxidative stress, hypoxia, and loss of adhesion, this result raises the possibility that the observed responses at T2 and T3 reflect transient stress responses specific to the protocol rather than essential steps in adaptation.

Overall, this study provides important insight into the transcriptional dynamics underlying suspension adaptation, but also highlights key gaps in mechanistic understanding, validation, and scalability. Addressing these limitations, particularly through validation of transient responses, testing in serum-free conditions, and evaluating transferability across cell types will be critical for developing scalable, reproducible suspension culture systems, advancing the commercial viability of cellular agriculture.

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